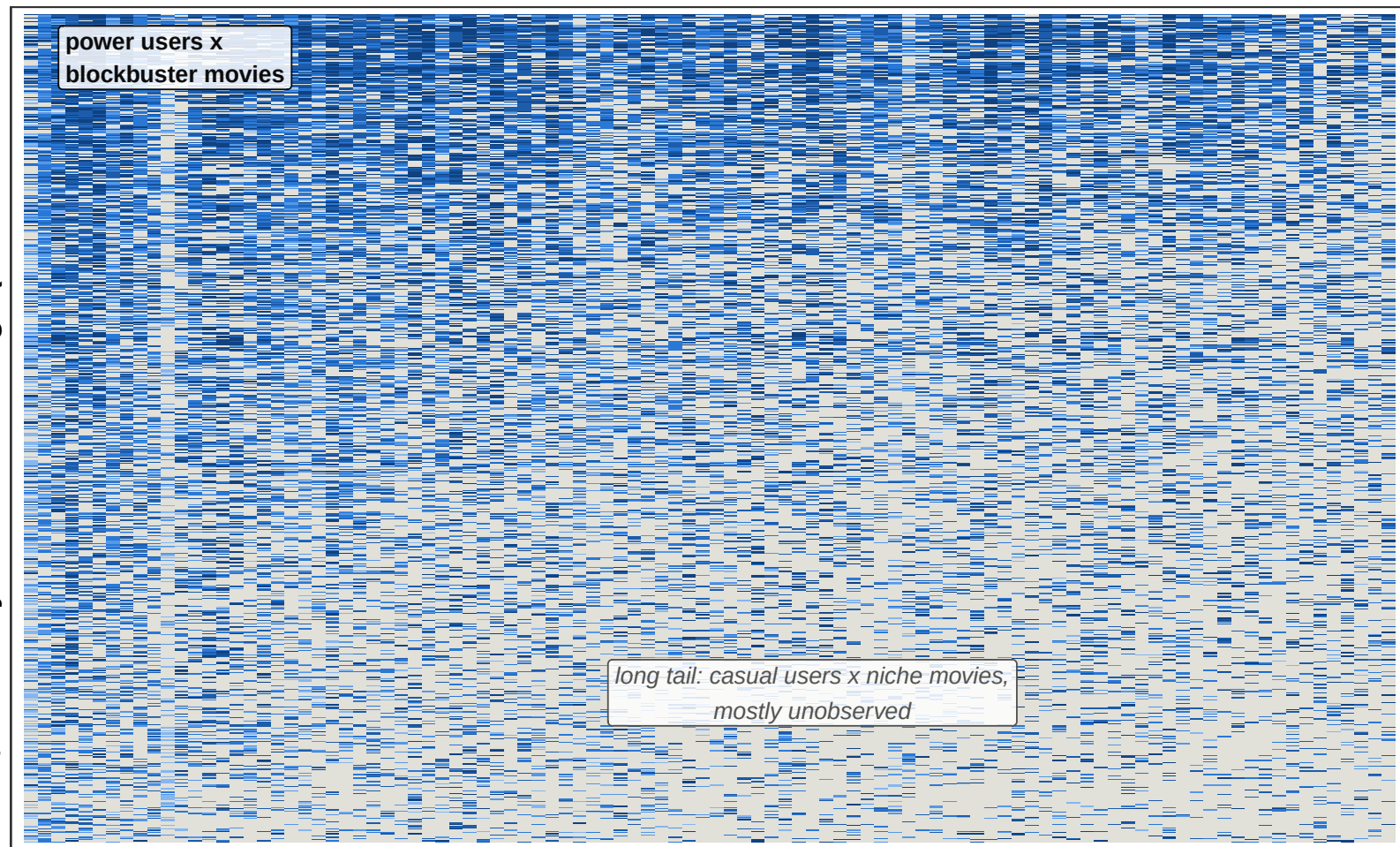


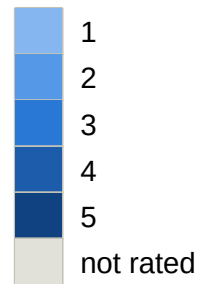
A recommender system's ratings matrix is mostly missing

MovieLens 100k, 100 most active users x 700 most-rated movies – only 39.5% observed

Movies, sorted by number of ratings (most rated – least)



Rating



Users, sorted by number of ratings (most active – least)